

Alexander Reinthal

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Cybersecurity ML engineer moving into frontier AI safety. 7 years building classifiers and detection pipelines for malicious traffic and software vulnerabilities (NTT Security; peer-reviewed exploit-prediction methodology adopted by Recorded Future). Recent AI safety work: 1st place ARENA 7.0 hackathon on deception probes, 4th of 641 at Apart Research on cross-request misuse detection in model APIs, and experience from AI control research at AI Safety Camp.

Publications

Data Modeling for Predicting Software Exploits

Nov 2018

Reinthal, A., Filippakis, E., Almgren, M.

[10.1007/978-3-030-03638-6_21](https://doi.org/10.1007/978-3-030-03638-6_21) (Springer LNCS: Nordic Conference on Secure IT Systems)

Research Projects

[Detecting Piecewise Cyber Espionage in Model APIs](#)

Nov 2025 – Nov 2025

Apart Research sprint, 4th of 641 submissions. Cross-context monitoring across model-API user sessions — directly analogous to detecting coordinated harms in production frontier deployments.

- Showed that piecewise misuse — individual requests that look benign in isolation — can be detected by modelling activity across the cyber kill chain.
- Aggregated signals across disparate requests (e.g. IP correlation) to surface attack patterns that per-request guardrails like Llama Guard 3 miss.
- Co-authored with Arthur Colle, David Williams-King, Yingquan Li, and Lihn Le.

Novel AI Control Protocol Classes: Evaluation and Scalability

Jan 2026 – present

AI Safety Camp research. Studied whether hierarchical and parallel control structures offer better safety-usefulness Pareto frontiers than simple trusted-untrusted pairs as the capability gap widens.

- Owned red-teaming work-stream: identified which open-weight models best bypass refusal training to use as untrusted models in control evals.
- Relevant to DeepMind responsibilities around AI control, agentic threat modelling, and long-horizon attack robustness.

[Detecting Deception in Chinese Models](#)

Jan 2026 – present

ARENA 7.0 weekend hackathon, 1st place. Linear probes for deception and politically sensitive lying across frontier open-weight models.

- Linear probes for deception transfer to Llama-3.3-70B (AUROC > 0.74) but fail on Qwen-2.5-72B (AUROC < 0.5) while Qwen is more China-aligned on evals than Llama-3.3. A curious gap that warrants further research. Ongoing.
- Awarded a Bluedot Impact rapid grant to extend the work to additional frontier open-weight models.

[About Emergent Misalignment](#)

Jan 2026 – present

ARENA 7.0 capstone, 3rd place. SFT-based study of how emergent misalignment couples to capability degradation.

- Fine-tuned model organisms via SFT; showed emergent misalignment degrades capabilities by 68pp on GSM8K and that any correct-data SFT simultaneously restores capabilities and reduces EM — effects are inseparable under this training recipe.

Experience

Facilitator

Mar 2026 – present

Bluedot Impact

- Part-time facilitator and mentor for the Technical AI Safety course and project track; consistent outstanding feedback from course participants.

Alum

Jan 2026 – Feb 2026

ARENA 7.0

- Research projects placed 1st (Detecting Deception in Chinese Models) and 3rd (Inoculation Prompting Against Emergent Misalignment).
- Five-week intensive curriculum on transformer internals, mechanistic interpretability, RLHF, evals, JAX, and the Inspect framework — taught by leading AI safety researchers.

Data Platform Engineer / Tech Lead / Data Scientist

Gothenburg, Sweden

Knowit Solutions Cocreate

Feb 2022 – present

- Implemented an OAuth-authenticated MCP gateway to secure agentic workflows — controlling tool access and authenticating model-side calls against downstream APIs.
- Compared LLM observability tooling (LangChain, Langfuse, Arize Phoenix) for a job-ad-to-consultant matching platform; deployed Arize Phoenix to production for LLM-call monitoring and model-performance tracking. Mentored junior engineers.

Security Analyst

Gothenburg, Sweden

NTT Security

Apr 2019 – Jan 2022

- Built ML classifiers and anomaly-detection pipelines for malicious network traffic across high-volume customer environments using Python and distributed processing.
- Designed alert-triage automation; mentored junior analysts on threat intelligence, anomaly detection, and triage workflows.

Python Software Engineer

Gothenburg, Sweden

Ericsson

June 2017 – Oct 2018

- Led a team building a log-parsing and analytics tool that grew into a dedicated Ericsson engineering team.

Open Source and Personal Projects

[Scheming Propensity Evals on Gemini 3.1 Pro](#)

May 2026 – present

Preparing scheming-propensity evaluations for upstreaming to Inspect Framework, following the methodology from [arXiv:2603.01608](#).

- Implemented the eval suite in Inspect and validated it against Gemini 3.1 Pro. Used to establish baselines on open-source models for model organisms research.

[Casually Jailbreaking Gemini 2.5 Flash](#)

2026 – 2026

Jailbroke Gemini 2.5 Flash with a few-shot in-context continuation attack, eliciting harmful recommendations.

- Practical demonstration of adversarial-robustness gaps in a production frontier model — directly relevant to the team's automated red-teaming and multi-turn attack work.

Open Source Contributions

Data-engineering contributions: [Dagster PR #24188](#) (ML pipeline orchestration) and [DLT Hub PR #594](#) (ingestion reliability). Full activity: github.com/reinthal.

Skills

Machine Learning & AI: PyTorch, JAX (learned during ARENA 7.0), Einops, XGBoost, SciKit Learn, Jupyter; LLM evals & observability (Inspect, LangChain, Langfuse, Arize Phoenix); linear probes, SFT, automated red-teaming

Programming Languages: Python, JavaScript, SQL, Rust, C/C++, R, Nix, Terraform

Platforms & Tools: Modal, Runpod, Openweights, Weights & Biases, Spark, Flink, Iceberg, Databricks, Dagster, Snowflake, dbt, Apache Superset

Infrastructure & DevOps: Nix, Docker, Kubernetes, FluxCD, GitOps, CI/CD pipelines, Terraform, AWS (EC2, S3, IAM)

Certifications: Bluedot Impact — Technical AI Safety (2026-03); AGI Strategy (2025-10)

Education

Chalmers University of Technology

MS in Engineering Physics

Gothenburg, Sweden

Sept 2016 – June 2018

- Coursework: Statistical Physics, Neural Networks and Machine Learning

Gothenburg University

BS in Computer Science

Gothenburg, Sweden

Sept 2013 – June 2016

- Coursework: Algorithms, Testing/Debugging/Verification, Theoretical Computer Science, Operating Systems, Cryptography, Cyber Security